Build: Modal / Dialog Component

2 min read Updated 23 Apr 2026 Topic 58 of 62

Problem: Build a reusable modal with overlay, close on escape key, and close on outside click.

What This Tests: Portals, event listeners, cleanup, the children prop, focus management.

jsx

JSX Copy

```jsx
function Modal({ isOpen, onClose, title, children }) {
  const modalRef = useRef(null);

  useEffect(() => {
    const handleEscape = (e) => {
      if (e.key === 'Escape') onClose();
    };

    if (isOpen) {
      document.addEventListener('keydown', handleEscape);
      document.body.style.overflow = 'hidden'; // Prevent background scrol
    }

    return () => {
      document.removeEventListener('keydown', handleEscape);
      document.body.style.overflow = '';
    };
  }, [isOpen, onClose]);

  const handleOverlayClick = (e) => {
    if (e.target === e.currentTarget) onClose(); // Only close if clicking
  };

  if (!isOpen) return null;
```

```
      position: 'fixed', top: 0, left: 0, right: 0, bottom: 0,
      backgroundColor: 'rgba(0,0,0,0.5)',
      display: 'flex', alignItems: 'center', justifyContent: 'center',
      zIndex: 1000,
    }}>
      <div ref={modalRef} style={{
        background: 'white', borderRadius: 8, padding: 24,
        minWidth: 320, maxWidth: 500, position: 'relative',
      }}>
        <div style={{ display: 'flex', justifyContent: 'space-between', al
          <h2 style={{ margin: 0 }}>{title}</h2>
          <button onClick={onClose} style={{ border: 'none', background: '
        </div>
        {children}
      </div>
    </div>,
    document.body
  );
}

// Usage
function App() {
  const [showModal, setShowModal] = useState(false);

  return (
    <div>
      <button onClick={() => setShowModal(true)}>Open Modal</button>
      <Modal isOpen={showModal} onClose={() => setShowModal(false)} title=
        <p>Are you sure you want to proceed?</p>
        <button onClick={() => setShowModal(false)}>Cancel</button>
        <button onClick={() => { alert('Confirmed!'); setShowModal(false);
      </Modal>
    </div>
  );
}
```

Key Concepts: `createPortal` for rendering outside the DOM hierarchy, Escape key listener with cleanup, overlay click detection with `e.target === e.currentTarget`, scroll lock on body, reusable via children prop.

1. When <Modal isOpen /> is rendered from inside App, where does the modal's markup actually end up in the DOM?

- ◯ Inside App's own container div, exactly where <Modal /> sits in the JSX
- ◯ Inside a shadow DOM root attached to the element held by modalRef
- ◯ Appended to document.body, outside the parent component's DOM hierarchy
- ◯ Inside a #modal-root element that React creates automatically for portals

Check answer

2. The overlay's click handler runs `if (e.target === e.currentTarget) onClose()`. Why is that comparison needed?

- ◯ Clicks inside the dialog bubble up to the overlay, so the check ensures only clicks landing on the overlay itself close the modal
- ◯ It makes onClose fire only on the very first click and be ignored on later clicks
- ◯ It stops the click from bubbling further up to document and firing onClose a second time
- ◯ Portal events skip React's synthetic event system, so target must be compared manually

Check answer

3. The effect depends on [isOpen, onClose]. When the modal goes from open to closed, what does React do with that effect?

- ◯ It removes the keydown listener but leaves document.body.style.overflow as 'hidden' until unmount

It re-runs the effect body and attaches a second keydown listener because isOpen changed

It runs the cleanup, removing the keydown listener and resetting document.body.style.overflow to ''

Nothing runs until Modal unmounts, so the keydown listener stays attached while the modal is closed

Check answer

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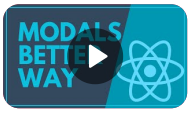
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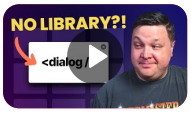
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Hand-picked references for this lesson.